MOOC Econometrics — Case Project: House Prices

Final Submission Answer Sheet

Dataset: Anglin & Gencay (1996) Hedonic Price Function (N = 546)

Question (a): Ramsey RESET Test for Linearity

Objective: Test whether the relationship between log(sell) and log(size/lot) is linear for houses not near the beach (d_beach = 0).

Methodology & Hypothesis:
1. Estimate the Restricted Baseline Model:
$\log(sell) = \beta_0 + \beta_1 \log(size) + \varepsilon$
2. Obtain the fitted values (y_hat).
3. Estimate the Unrestricted Auxiliary Model by introducing higher-order polynomial terms:
$\log(sell) = \beta_0 + \beta_1 \log(size) + \gamma_1 \hat{y}^2 + \gamma_2 \hat{y}^3 + \varepsilon$

Hypothesis Testing:
H_0: $\gamma_1 = \gamma_2 = 0$ (The linear specification is correct)
H_1: $\gamma_1 \neq 0$ or $\gamma_2 \neq 0$ (The relationship is non-linear)

An F-test is conducted to compare the sum of squared residuals between both models. If the p-value < 0.05, we reject the null hypothesis of linearity.

Question (b): Functional Form Selection (Linear vs. Log-Linear)

Analysis: Using the logarithm of the dependent variable log(sell) is statistically superior to using raw house price (sell) for three primary reasons:

1. **Mitigation of Heteroskedasticity:** House prices exhibit larger variances as size increases. Log transformation stabilizes this variance.
2. **Normality of Residuals:** The raw pricing data is heavily right-skewed. Taking logs normalizes the distribution, ensuring valid OLS inference and t-tests.
3. **Multiplicative Interpretation:** Real estate characteristics (e.g., adding a bathroom) typically scale house values by a percentage increment rather than a fixed monetary value.

Question (c): Hedonic Pricing Model Specification

The full multiple linear regression specification incorporating structural, aesthetic, and location characteristics is modeled as follows:

$$\log(sell) = \beta_0 + \beta_1 \log(size) + \beta_2 bdms + \beta_3 fb + \beta_4 sty + \beta_5 drv + \beta_6 rec + \beta_7 ffn + \beta_8 gwh + \beta_9 ac + \beta_{10} gar + \beta_{11} reg + \varepsilon$$

Where 'size' represents the lot size, variables like 'bdms' (bedrooms), 'fb' (full bathrooms), 'sty' (stories), and 'gar' (garage places) are discrete structural values, and the remaining variables represent binary structural/locational dummy features (1 if yes, 0 if no).

Question (d): Interpretation of Dummy Variables

For log-linear models, the coefficient (β) of a dummy variable indicates the relative effect of that characteristic on house price. Since dummy variables reflect discrete switches (from 0 to 1), the exact percentage change in the selling price is computed as:

$$\% \Delta Price = 100 \times (e^{\beta} - 1)$$

For instance, if the estimated coefficient for air conditioning (β_{ac}) is 0.15, the actual expected percentage increase in price for a house with AC compared to one without AC, keeping all other variables constant, is $100 \times (e^{0.15} - 1) \approx 16.18\%$.

Question (e): Diagnostic Testing for Heteroskedasticity

Breusch-Pagan Test: OLS assumes constant error variance (homoskedasticity). To test this, we regress the squared residuals (ε^2) from the main model onto all independent explanatory variables.

H_0: $\sigma_i^2 = \sigma^2$ (Errors are homoskedastic)
H_1: Variance depends on the independent variables

If heteroskedasticity is detected (p-value < 0.05), standard errors must be corrected using White's heteroskedasticity-robust standard errors to ensure reliable significance tests.

Appendix: R Studio Automation Script Template

```
# Load required libraries
library(lmtest)
library(sandwich)

# 1. Load Data
data <- read.csv('HousePrices.csv') # Ensure correct file path

# 2. Part (a): Ramsey RESET Test
sub_data <- subset(data, d_beach == 0)
model_a <- lm(log(sell) ~ log(size), data = sub_data)
reset_result <- resettest(model_a, power = 2:3, type = 'fitted')
print(reset_result)

# 3. Part (c): Hedonic Regression Model
full_model <- lm(log(sell) ~ log(size) + bdms + fb + sty + drv + rec + ffn +
gwh + ac + gar + reg, data = data)
summary(full_model)

# 4. Part (e): Breusch-Pagan Test
bp_result <- bptest(full_model)
print(bp_result)
```

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